Electronic Devices and Circuits

Topics to know and revise before the tests · English version

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1. Fundamental electrical quantities and circuit laws

- Voltage, current, resistance and electrical power
- Ohm's law
- voltage/current directions and sign conventions
- Kirchhoff's Current Law (KCL)
- Kirchhoff's Voltage Law (KVL)
- series and parallel resistors
- equivalent resistance
- voltage divider
- current divider
- practical voltage and current sources and internal resistance

2. Methods for analysing linear electrical circuits

- Superposition principle
- deactivation of independent voltage and current sources
- Thévenin's theorem
- determination of Thévenin voltage and resistance
- Norton's theorem
- Thévenin–Norton source conversion
- equivalent circuits for load analysis
- maximum power transfer condition

3. Periodic signals and frequency-domain representation

- Amplitude, frequency, angular frequency and phase
- DC component
- decomposition into sinusoidal components
- basic idea of Fourier decomposition
- amplitude spectrum and phase spectrum
- interpretation of multi-frequency signals

4. R, L and C elements in AC circuits

- Voltage–current relations for R, C and L
- phase relations
- complex/phasor representation
- impedance and admittance
- impedance of resistor, capacitor and inductor
- real and imaginary parts of impedance
- low- and high-frequency behaviour of capacitors and inductors
- impedance of RC, RL and RLC networks

5. Resonance and RLC circuits

- Series and parallel resonance
- resonant frequency
- cancellation of capacitive and inductive reactances
- impedance versus frequency

- resonance curves
- bandwidth and −3 dB points
- influence of losses/resistance on resonance

6. Passive filters and frequency characteristics

- Transfer function
- magnitude and phase response
- RC low-pass filter
- CR high-pass filter
- cut-off frequency
- LC/RLC filters
- band-pass response
- notch/band-stop response
- decibels
- Bode magnitude and phase plots
- asymptotic slopes such as ±20 dB/decade and ±40 dB/decade

7. Fundamentals of semiconductor physics

- Atomic structure and valence electrons
- Pauli exclusion principle and discrete energy states
- energy bands in solids
- valence and conduction bands
- band gap
- conductors, semiconductors and insulators in the energy-band picture
- band overlapping and touching
- direct and indirect band-gap semiconductors
- basic properties of Si, Ge and compound semiconductors

8. Charge carriers in semiconductors

- Electrons and holes
- electron–hole pair generation
- recombination
- intrinsic semiconductor
- intrinsic carrier concentration
- temperature dependence of carrier concentration
- electron and hole mobility
- drift current
- electric field and drift velocity
- conductivity and resistivity
- temperature dependence of mobility and resistivity

9. Doped semiconductors

- Donor and acceptor impurities
- n-type and p-type material
- majority and minority carriers
- mass-action law ($np = n_i^2$)
- influence of doping on resistivity
- dopant ionisation
- freeze-out, extrinsic and intrinsic temperature regions
- Fermi level
- Fermi–Dirac distribution
- Fermi-level position in intrinsic, n-type and p-type material

10. Basic semiconductor fabrication techniques

- Epitaxy
- diffusion
- dopant concentration profiles
- ion implantation
- silicon oxidation and SiO_2 layers
- masking and photolithography
- local modification of semiconductor properties
- general sequence of silicon-device fabrication

11. The p–n junction

- Formation of the p–n junction
- carrier diffusion
- depletion region
- space charge and built-in electric field
- potential barrier
- energy-band representation
- unbiased junction
- forward-biased p–n junction
- reverse-biased p–n junction
- diffusion and drift currents
- influence of external voltage on depletion-region width

12. Diode current–voltage characteristic

- Shockley diode equation
- saturation current
- thermal voltage
- ideality/emission factor
- forward characteristic
- reverse leakage current
- static and differential/dynamic resistance
- ideal versus real diode

13. Diode models for circuit analysis

- Ideal-diode model
- constant-voltage-drop model
- piecewise-linear model
- series resistance
- determining whether a diode is ON or OFF
- analysis of circuits containing multiple diodes

14. p–n and Schottky diodes

- Metal–semiconductor junction
- Schottky-diode operating principle
- forward-voltage differences between Si p–n and Schottky diodes
- reverse leakage
- switching speed
- main differences between Schottky and p–n diodes
- basic construction of small-signal and power diodes

15. Dynamic behaviour of diodes

- Charge storage

- transition from forward to reverse bias
- reverse-recovery time (t_{rr})
- reverse-recovery current
- influence of diode technology on switching speed
- dynamic effects in switching circuits

16. Diode maximum ratings

- Maximum forward current
- maximum reverse voltage
- maximum power dissipation
- allowable pulse current/energy
- consequences of exceeding maximum ratings

17. Thermal behaviour of diodes

- Temperature dependence of saturation current
- increase of reverse current with temperature
- temperature shift of the forward I–V characteristic
- negative temperature coefficient of Si forward voltage (approximately -1.7 to -2 mV/K at constant current)
- physical origin of diode temperature dependence

18. Junction breakdown and Zener diodes

- Zener breakdown
- avalanche breakdown
- relation between doping and breakdown mechanism
- differences between Zener and avalanche mechanisms
- Zener-diode characteristic
- Zener voltage
- dynamic resistance in breakdown
- voltage stabilisation/limiting using a Zener diode

19. Junction capacitance and varactor diode

- Depletion region as a dielectric
- p–n junction capacitance
- capacitance versus reverse voltage
- varicap/varactor diode
- voltage-controlled capacitance

20. BJT construction and operating principle

- NPN and PNP structures
- emitter, base and collector
- two p–n junctions
- carrier transport through the base
- emitter, base and collector currents
- $I_E = I_C + I_B$
- current gain β
- physical origin of transistor action

21. BJT characteristics and operating regions

- Input characteristic
- $I_C(U_{CE})$ output characteristics
- dependence of I_C on I_B and U_{BE}
- cut-off
- forward-active region

- saturation
- breakdown
- Early effect
- Early voltage
- finite output resistance
- variation of current gain β

22. BJT operating point

- Transistor biasing
- Q-point
- DC load line
- intersection of load line and device characteristic
- selection of I_C and U_CE
- operating point versus available signal amplitude
- clipping/distortion caused by saturation and cut-off

23. BJT small-signal model

- Linearisation around operating point
- small changes of currents and voltages
- h-parameters
- input resistance
- output conductance/resistance
- small-signal current gain
- application to amplifier analysis

24. BJT maximum ratings and thermal behaviour

- Maximum junction and collector–emitter voltages
- breakdown
- maximum collector current
- maximum power dissipation
- safe operating area
- short-duration overloads
- junction temperature
- junction–case–heatsink–ambient thermal resistance
- calculation of junction temperature
- temperature dependence of U_BE, I_C and β
- thermal stability of the operating point

25. Basic BJT amplifier configurations

- Common emitter (CE)
- common collector (CC, emitter follower)
- common base (CB)
- voltage and current gain
- input and output resistance
- phase inversion in CE
- comparison of CE, CB and CC
- emitter resistance as local negative feedback
- distortion reduction and gain stabilisation
- internal capacitances and Miller effect at high frequencies

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26. Types of field-effect transistors

- Basic field-effect control principle
- JFET and MOSFET
- n-channel and p-channel devices
- enhancement-mode MOSFET
- depletion-mode MOSFET
- gate, drain and source
- U_GS and U_DS

27. MOSFET and JFET construction and operation

- MOS structure
- insulated gate and oxide layer
- channel formation and disappearance
- threshold voltage U_GS(th)
- control of channel conductivity by gate voltage
- depletion region in a JFET
- channel pinch-off
- comparison of JFET and MOSFET

28. FET characteristics and operating regions

- I_D(U_DS) output characteristics
- I_D(U_GS) transfer characteristic
- cut-off region
- linear/ohmic region
- saturation region
- boundaries between operating regions
- drain-current equations
- FET used as a voltage-controlled resistor

29. FET non-idealities and small-signal model

- Channel-length modulation / FET equivalent of the Early effect
- finite output resistance
- transconductance g_m
- output conductance g_ds
- small-signal equivalent model
- linearisation around Q-point

30. FET biasing and amplifier configurations

- Operating-point determination
- load line
- common-source amplifier
- voltage gain
- very high gate input resistance
- common-drain/source-follower stage
- input and output resistances

31. FET maximum ratings and temperature effects

- Maximum gate–source voltage
- gate-oxide/gate breakdown
- drain–source breakdown

- maximum drain current
- maximum power dissipation
- minimum R_DS(on)
- safe operating area
- temperature dependence of threshold voltage and channel parameters

32. Basic photometric quantities

- Optical radiation and spectral sensitivity of the human eye
- luminous flux — lumen
- luminous intensity — candela
- solid angle — steradian
- illuminance — lux
- geometrical relations for illuminance
- spectral luminous efficiency

33. Photoelectric effects in semiconductors

- Photon energy $E = hf = hc/\lambda$
- optical generation of electron–hole pairs
- band gap and maximum detectable wavelength
- spectral response versus semiconductor material
- limitations in UV, visible and IR detection

34. Photoresistors

- Operating principle
- resistance versus illumination
- nonlinear response
- temperature dependence
- limited response speed

35. Photodiodes and phototransistors

- Generation of photocurrent
- reverse-biased photodiode operation
- I–V characteristic under illumination
- spectral sensitivity
- response speed
- phototransistor principle
- internal current gain
- sensitivity–speed trade-off

36. Optocouplers

- LED and phototransistor in a single component
- galvanic isolation
- current transfer ratio (CTR)
- applications of optical isolation
- bandwidth/speed limitations

37. Light-emitting diodes

- Radiative recombination
- direct-band-gap materials
- relation between band gap and emitted wavelength
- LED materials such as GaAs, GaAsP, GaP and GaN
- LED I–V characteristic
- driving an LED from a voltage source

- current-limiting resistor
- electrical-to-optical efficiency
- forward voltage versus LED material/colour

38. CCD and CMOS image sensors

- Photogeneration and charge accumulation in a potential well
- pixel structure
- charge transfer and readout in CCD sensors
- basic CCD–CMOS differences
- analogue pixel readout
- correlated double sampling (CDS)

39. Operational-amplifier fundamentals

- Inverting and non-inverting inputs
- differential input voltage
- open-loop voltage gain
- output saturation
- input and output resistance
- ideal op-amp model
- infinite gain, infinite input resistance, zero output resistance and zero input current assumptions

40. Negative feedback

- Open-loop and closed-loop systems
- feedback factor β
- closed-loop gain
- gain stabilisation by negative feedback
- virtual short between op-amp inputs under correct negative-feedback operation
- influence of feedback on circuit impedances

41. Basic op-amp circuits

- Voltage follower
- non-inverting amplifier
- inverting amplifier
- determination of gain from resistor values
- input resistance
- signal-source buffering
- practical use of ideal-op-amp assumptions

42. Op-amp bandwidth and stability

- Frequency-dependent open-loop gain
- transfer-function poles
- gain and phase Bode plots
- feedback at high frequencies
- transition from negative to positive feedback
- oscillation condition
- phase margin
- gain margin
- frequency compensation
- dominant pole
- gain–bandwidth product

43. Op-amp non-idealities

- Finite gain

- finite bandwidth
- gain–bandwidth limitation
- slew rate
- large-signal distortion caused by slew-rate limitation
- common-mode and differential signals
- differential gain
- common-mode gain
- CMRR and expression in dB

44. Conditions for sinusoidal oscillation

- Oscillator as amplifier with feedback
- loop gain
- amplitude condition
- phase condition
- Barkhausen criterion
- start-up condition
- steady-state amplitude
- nonlinearities in amplitude stabilisation

45. RC oscillators

- Wien-bridge oscillator
- Wien-bridge oscillation frequency
- required amplifier gain
- RC phase-shift oscillator
- cascaded RC phase-shifting networks
- phase and gain requirements
- nonlinear/diode amplitude stabilisation

46. LC oscillators

- LC resonator as frequency-selective element
- phase and gain conditions
- Colpitts oscillator
- Hartley oscillator
- oscillation frequency from L and C
- required feedback-element ratios

47. Crystal oscillators

- Quartz resonator
- piezoelectric effect
- equivalent electrical model
- series resonance
- parallel resonance
- small separation between series and parallel resonance
- high Q and frequency stability
- advantages over simple LC oscillators

48. Relaxation and non-sinusoidal oscillators

- Relaxation-oscillator principle
- capacitor charging and discharging
- switching thresholds
- op-amp astable multivibrator
- positive feedback and hysteresis

- square-wave generation
- period and frequency as functions of R, C and switching thresholds

49. Variable-duty-cycle generators

- Separate capacitor charging and discharging paths
- diodes to obtain different time constants
- high-state and low-state durations
- duty cycle
- generation of non-50% duty-cycle waveforms

50. Square/triangle-wave generator

- Comparator/Schmitt trigger producing the square wave
- integrator producing triangular wave
- ramp slope as function of voltage, R and C
- comparator thresholds versus triangle-wave amplitude
- oscillation period and frequency